

SP Assignment 1

1. opgave

- a) ideen er at bruge uafhængigheden samt at Y_i erne er kendte givet filteret for $n \in \mathbb{N}$

$$\begin{aligned} E[e^{\alpha S_{n+1}} | \mathcal{F}_n] &= E\left[e^{\alpha \sum_{i=1}^{n+1} Y_i} | \mathcal{F}_n\right] = E\left[\prod_{i=1}^{n+1} e^{\alpha Y_i} | \mathcal{F}_n\right] = \prod_{i=1}^n e^{\alpha Y_i} \cdot E[e^{\alpha Y_{n+1}} | \mathcal{F}_n] \\ &= \prod_{i=1}^n e^{\alpha Y_i} \cdot E[e^{\alpha Y_{n+1}}] = e^{\alpha S_n} \cdot E[e^{\alpha Y_{n+1}}] = e^{\alpha S_n} \cdot E[e^{\alpha Y_1}] \end{aligned}$$

3. lighed er sand siden at Y_i 'erne er kendte i filteret

4. er fordi at $Y_{n+1} \perp \mathcal{F}_n$

For $n = 0$

$$E[e^{\alpha Y_1} | \mathcal{F}_0] = E[e^{\alpha Y_1}] = e^{\alpha S_0} \cdot E[e^{\alpha Y_1}]$$

Dette har vi vist til forelæsningen

Så vi vil egentlig bare gerne regne $E[e^{\alpha Y_1}]$ da $Y_1 =^d Y_n \forall n \in \mathbb{N}$

Vi har vis

Definitionen:

$$= \int_{\mathbb{R}} e^{\alpha y} P_{Y_1}(dy) = \int_{\mathbb{R}} e^{\alpha y} \left(e^{-2y} \cdot \mathbb{I}_{\{\geq 0\}}(y) + \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{\{< 0\}}(y) \right) \lambda(dy)$$

Positive funktioner

$$= \int_{\mathbb{R}} e^{\alpha y} \cdot e^{-2y} \cdot \mathbb{I}_{\{\geq 0\}}(y) \lambda(dy) + \int_{\mathbb{R}} e^{\alpha y} \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{\{< 0\}}(y) \lambda(dy)$$

1. integral:

monoton konvergens

$$\int_{\mathbb{R}} e^{\alpha y} \cdot e^{-2y} \cdot \mathbb{I}_{\{\geq 0\}}(y) \lambda(dy) = \lim_{n \rightarrow \infty} R \int_0^n e^{y(\alpha-2)} dy = \lim$$

Forskellige værdier af α

$\alpha = 2$:

$$\lim_{n \rightarrow \infty} R \int_0^n e^{y(\alpha-2)} dy = \lim_{n \rightarrow \infty} \int_0^n 1 = \infty$$

$\alpha \neq 2$:

$$\lim_{n \rightarrow \infty} R \int_0^n e^{y(\alpha-2)} dy = \lim_{n \rightarrow \infty} \left[\frac{e^{y(\alpha-2)}}{\alpha-2} \right]_0^n = \frac{e^{n(\alpha-2)} - 1}{\alpha-2}$$

$\alpha > 2$:

$$\frac{e^{n(\alpha-2)} - 1}{\alpha-2} \rightarrow \infty \text{ for } n \rightarrow \infty$$

$\alpha < 2$:

$$\frac{e^{n(\alpha-2)} - 1}{\alpha-2} \rightarrow \frac{1}{2-\alpha} \text{ for } n \rightarrow \infty$$

$$\int_{\mathbb{R}} e^{\alpha y} \cdot e^{-2y} \cdot \mathbb{I}_{\{\geq 0\}}(y) \lambda(dy) = \begin{cases} \frac{1}{2-\alpha} & \text{for } \alpha < 2 \\ \infty & \text{for } \alpha \geq 2 \end{cases}$$

2. integral:

$$\int_{\mathbb{R}} e^{\alpha y} \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{\{<0\}}(y) \lambda(dy)$$

Igen monoton konvergens

$$\int_{\mathbb{R}} e^{\alpha y} \frac{1}{2} \cdot e^y \cdot \frac{1}{2} \cdot \mathbb{I}_{\{<0\}}(y) \lambda(dy) = \frac{1}{2} \lim_{n \rightarrow \infty} R \int_{-n}^0 e^{y(\alpha+1)} dy$$

Igen 3 cases:

$\alpha < -1$:

$$\lim_{n \rightarrow \infty} R \int_{-n}^0 e^{y(\alpha+1)} dy = \lim_{n \rightarrow \infty} \left[\frac{e^{y(\alpha+1)}}{\alpha+1} \right]_{-n}^0 = \lim_{n \rightarrow \infty} \left(\frac{1}{\alpha+1} - \frac{e^{-n(\alpha+1)}}{\alpha+1} \right) = \infty$$

Da $\frac{e^{-n(\alpha+1)}}{\alpha+1} \rightarrow 0$

$\alpha = -1$:

$$\lim_{n \rightarrow \infty} R \int_{-n}^0 e^{y(\alpha+1)} dy = \lim_{n \rightarrow \infty} \int_{-1}^0 1 = \infty$$

$\alpha > -1$:

$$\lim_{n \rightarrow \infty} R \int_{-n}^0 e^{y(\alpha+1)} dy = \lim_{n \rightarrow \infty} \left[\frac{e^{y(\alpha+1)}}{\alpha+1} \right]_{-n}^0 = \lim_{n \rightarrow \infty} \left(\frac{1}{\alpha+1} - \frac{e^{-n(\alpha+1)}}{\alpha+1} \right) = \frac{1}{\alpha+1}$$

Altså

$$\int_{\mathbb{R}} e^{\alpha y} \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{\{<0\}}(y) \lambda(dy) = \begin{cases} \infty & \text{for } \alpha \leq -1 \\ \frac{1}{2} \cdot \frac{1}{\alpha+1} & \text{for } \alpha > -1 \end{cases}$$

Så hvis vi summer får vi

$$E[e^{\alpha Y_1}] = \begin{cases} \infty & \text{for } \alpha \leq -1 \\ \infty & \text{for } \alpha \geq 2 \\ \frac{1}{2} \frac{1}{\alpha+1} + \frac{1}{2-\alpha} & \text{for } \alpha \in (-1, 2) \end{cases}$$

Derived

$$E[e^{\alpha S_{n+1}} | \mathcal{F}_n] = e^{\alpha S_n} \cdot E[e^{\alpha Y_1}] = \begin{cases} e^{\alpha S_n} \cdot \left(\frac{1}{2} \frac{1}{\alpha+1} + \frac{1}{2-\alpha} \right) & \text{for } \alpha \in (-1, 2) \\ \infty & \text{ellers} \end{cases}$$

$\forall n \in \mathbb{N}_0$

b) calculate

$$\mathbb{P}(S_n < 0 < S_{n+1} | \mathcal{F}_n) \text{ for } n \in \mathbb{N}$$

Lad $n \in \mathbb{N}$

$$\begin{aligned} \mathbb{P}(S_n < 0 < S_{n+1} | \mathcal{F}_n) &= E[\mathbb{I}_{\{S_n < 0 < S_{n+1}\}} | \mathcal{F}_n] \\ &= E[\mathbb{I}_{\{S_n < 0\}} \cdot \mathbb{I}_{\{0 < S_{n+1}\}} | \mathcal{F}_n] = E\left[\mathbb{I}_{\{\sum_{i=1}^n Y_i < 0\}} \mathbb{I}_{\{-\sum_{i=1}^n Y_i < Y_{n+1}\}} \middle| \mathcal{F}_n\right] \end{aligned}$$

Nyttig regel. Y_i er kendt for $i \leq n$

$$= E\left[\mathbb{I}_{\{\sum_{i=1}^n Y_i < 0\}} \mathbb{I}_{\{-\sum_{i=1}^n Y_i < Y_{n+1}\}} \middle| y_i = Y_i \text{ for } i \in \{1, \dots, n\}\right]$$

Definer $s = \sum_{i=1}^n Y_i$

2 cases

$s \geq 0$:

$$E \left[\mathbb{I}_{\{\sum_{i=1}^n y_i < 0\}} \mathbb{I}_{\{-\sum_{i=1}^n y_i < Y_{n+1}\}} \right] = E[0] = 0$$

$s < 0$:

$$\begin{aligned} E \left[\mathbb{I}_{\{\sum_{i=1}^n y_i < 0\}} \mathbb{I}_{\{-\sum_{i=1}^n y_i < Y_{n+1}\}} \right] &= E \left[\mathbb{I}_{\{-\sum_{i=1}^n y_i < Y_{n+1}\}} \right] = \int_{\mathbb{R}} \mathbb{I}_{(-s, \infty)}(y) P_{Y_1}(dy) \\ &= \int_{\mathbb{R}} \mathbb{I}_{(-s, \infty)}(y) \left(e^{-2y} \cdot \mathbb{I}_{\{\geq 0\}}(y) + \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{\{< 0\}}(y) \right) \lambda(dy) \\ &= \int_{\mathbb{R}} \mathbb{I}_{(-s, \infty)}(y) \cdot e^{-2y} \cdot \mathbb{I}_{[0, \infty)}(y) \lambda(dy) + \int_{\mathbb{R}} \mathbb{I}_{(-s, \infty)}(y) \cdot \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{(-\infty, 0)}(y) \lambda(dy) \end{aligned}$$

1. integral

$$\begin{aligned} \int_{\mathbb{R}} \mathbb{I}_{(-s, \infty)}(y) \cdot e^{-2y} \cdot \mathbb{I}_{[0, \infty)}(y) \lambda(dy) &= \lim_{n \rightarrow \infty} R \int_{-s}^n e^{-2y} dy = \lim_{n \rightarrow \infty} \left[-\frac{e^{-2y}}{2} \right]_{-s}^n = \lim_{n \rightarrow \infty} \frac{e^{2s}}{2} - \frac{e^{-2n}}{2} \\ &= \frac{e^{2s}}{2} \end{aligned}$$

2. integral

$$\int_{\mathbb{R}} \mathbb{I}_{(-s, \infty)}(y) \cdot \frac{1}{2} \cdot e^y \cdot \mathbb{I}_{(-\infty, 0)}(y) \lambda(dy) = \int_{\mathbb{R}} 0 \lambda(dy) = 0$$

Som sammen giver

$$\mathbb{P}(S_n < 0 < S_{n+1} | \mathcal{F}_n) = \begin{cases} \frac{e^{2S_n}}{2} & \text{for } S_n < 0 \\ 0 & \text{for } S_n \geq 0 \end{cases}$$

Opgave 2

a) argumenter for at markovkæden er irreducible
ved induktion

" $0 \leftrightarrow 1$ ": $p_{01} = p_0 > 0, p_{10} = 1 - p_1 > 0$ per antagelse, da følger direkte af 6.6.1(b)

Lad $n \in \mathbb{N}$

Antag at $i \leftrightarrow j$ for $i, j \leq n$

Vi viser " $n \leftrightarrow n+1$ ":

Samme resultat

$$\begin{aligned} p_{n,n+1} &= p_n > 0 \Rightarrow "n \rightarrow n+1" \\ p_{n+1,0} &= 1 - p_{n+1} > 0 \Rightarrow "n+1 \rightarrow 0" \end{aligned}$$

Lad $i \in \{0, \dots, n\}$

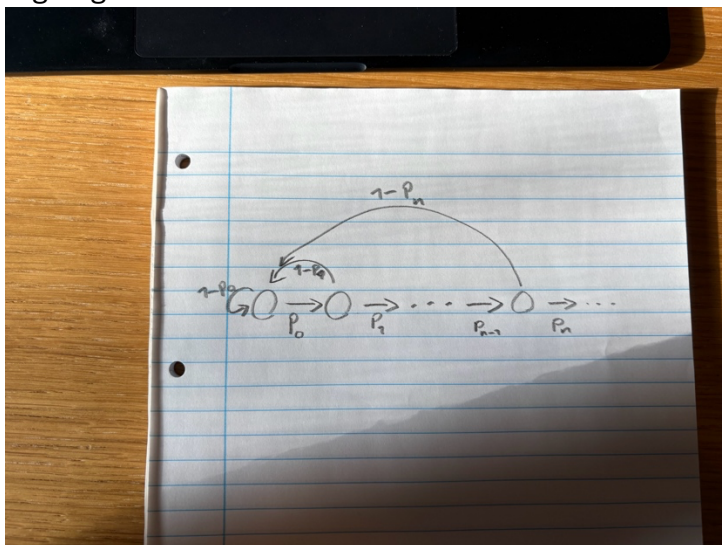
Per antagelse " $0 \rightarrow i$ " og " $i \rightarrow n$ "

men

$$\begin{aligned} n+1 \rightarrow 0, \quad 0 \rightarrow i &\Rightarrow n+1 \rightarrow i \\ i \rightarrow n, \quad n \rightarrow n+1 &\Rightarrow i \rightarrow n+1 \end{aligned}$$

altså $i \leftrightarrow n+1 \forall i \in \{0, \dots, n\}$

derud $i \leftrightarrow j \forall i, j \in S$ og markovkæden er irreducibel
tegning:



b) b

find p_{02}^4 vi ser at for at det skal kunne ske må vi skulle have $X_2 = 0, X_3 = 1, X_2 = 2$ altså

$$p_{02}^4 = p_{00}^2 \cdot p_0 \cdot p_1$$

Så vi skal bare have fundet p_{00}^2

$$\begin{aligned} p_{00}^2 &= \mathbb{P}_0(X_2 = 0) = \int_S \int_{\{0\}} Q(x_1, dx_2) Q(0, dx_1) \\ &= \sum_{i \in S} p_{0i} \cdot \left(\sum_{j \in S} p_{ij} \cdot \mathbb{I}_{\{0\}}(j) \right) = \sum_{i \in S} p_{0i} \cdot p_{i0} = p_{00} \cdot p_{00} + p_{01} \cdot p_{10} \\ &= (1 - p_0)^2 + p_0 \cdot (1 - p_1) = 1 - p_0 + p_0^2 - p_0 p_1 \\ p_{02}^4 &= p_{00}^2 \cdot p_0 \cdot p_1 = (1 - p_0 + p_0^2 - p_0 p_1) \cdot p_0 \cdot p_1 \end{aligned}$$

Herunder regner vi, hvor vi bruger at $\mathbb{P}_{X_{n-1}}(X_n \neq 0) = \mathbb{P}_{X_{n-1}}(X_n = X_{n-1} + 1) = p_{X_{n-1}}$

For $n = 1$

$$P_0(T_0 = 1) = p_{00} = 1 - p_0$$

For $n \geq 2$

$$\begin{aligned} P_0(T_0 = n) &= P_0(\{X_1 \neq 0, \dots, X_n = 0\}) = p_{n-1,0} \prod_{i=1}^{n-1} p_{i-1,i} = (1 - p_{n-1}) \prod_{i=0}^{n-2} p_i \\ P_0(T_0 = n) &= \begin{cases} 1 - p_0 & \text{for } n = 1 \\ (1 - p_{n-1}) \prod_{i=0}^{n-2} p_i & \text{for } n \geq 2 \end{cases} \end{aligned}$$

$$P_0(T_0 \geq n) = 1 - P_0(T_0 < n)$$

For $n = 1$

$$= 1 - P_0(T_0 < n) = 1$$

For $n \geq 2$

$$\begin{aligned} &= 1 - P_0\left(\bigcup_{i=1}^{n-1} \{T_0 = i\}\right) = 1 - \sum_{i=1}^{n-1} P_0(T_0 = i) = 1 - (1 - p_0) - \sum_{i=2}^{n-1} (1 - p_{i-1}) \cdot \prod_{k=0}^{i-2} p_k \\ &= p_0 - \sum_{i=2}^{n-1} (1 - p_{i-1}) \cdot \prod_{k=0}^{i-2} p_k = p_0 - \sum_{i=2}^{n-1} \left(\prod_{k=0}^{i-2} p_k - \prod_{k=0}^{i-1} p_k \right) \end{aligned}$$

Teleskoperende sum!

$$= p_0 - \left(p_0 - \prod_{k=0}^{i-2} p_k \right) = \prod_{k=0}^{i-2} p_k$$

Så

$$P_0(T_0 \geq n) = \begin{cases} 1 & \text{for } n = 1 \\ \prod_{k=0}^{i-2} p_k & \text{for } n \geq 2 \end{cases}$$

c) rekurens

$$0 \text{ er rekurent} \Leftrightarrow \mathbb{P}_0(T_0 < \infty) = 1 \Leftrightarrow \mathbb{P}_0(T_0 = \infty) = 0$$

$$\mathbb{P}_0(T_0 = \infty) = \mathbb{P}_0\left(\bigcap_{n=1}^{\infty} \{X_1 \neq 0, \dots, X_n \neq 0\}\right) = \lim_{n \rightarrow \infty} \mathbb{P}_0\left(\bigcap_{n=1}^N \{X_1 \neq 0, \dots, X_n \neq 0\}\right)$$

Den kan vi regne! Enda relativt nemt fordi der kun er "en sti"

$$\begin{aligned} \mathbb{P}_0\left(\bigcap_{n=1}^N \{X_1 \neq 0, \dots, X_n \neq 0\}\right) &= \mathbb{P}_0\left(\bigcap_{n=1}^N \{X_1 = 1, \dots, X_n = n\}\right) \\ \lim_{N \rightarrow \infty} \mathbb{P}_0\left(\bigcap_{n=1}^N \{X_1 = 1, \dots, X_n = n\}\right) &= \lim_{N \rightarrow \infty} \prod_{n=1}^N p_{n-1,n} = \prod_{n=0}^{\infty} p_n \end{aligned}$$

d) transiens

giv en følge af sandsynligheder så markovkæden er transient

$$p_i = 1 - \frac{1}{2^{i+1}}$$

Per del opgave (c)

$$0 \text{ transient} \Leftrightarrow \neg(0 \text{ recurrent}) \Leftrightarrow \prod_{n=0}^{\infty} p_n > 0$$

Da $1 - \frac{1}{2^{n+1}} \in (0,1)$ konvergerer produktet

$$\prod_{n=0}^N \left(1 - \frac{1}{2^{n+1}}\right)$$

Vi tager log på det endelige produkt for at tage exp på det senere når vi har vist at denne konvergerer

$$\log\left(\prod_{n=0}^N\left(1 - \frac{1}{2^{n+1}}\right)\right) = \sum_{n=0}^N \log\left(1 - \frac{1}{2^{n+1}}\right)$$

Den er monoton og vi bemærker at $-2x \leq \log(1-x)$ for $x \in (0, \frac{1}{2}]$

Derved

$$\sum_{n=0}^N \log\left(1 - \frac{1}{2^{n+1}}\right) \geq \sum_{n=0}^N -2 \cdot \frac{1}{2^{n+1}} = \sum_{n=0}^N -\frac{1}{2^n}$$

$$\lim_{N \rightarrow \infty} \sum_{n=0}^N -\frac{1}{2^n} = -2$$

Derved monoton og begrænset og så konvergent

Da exp er kontinuert har vi ligeså konvergens og følgende ulighed

$$\sum_{n=0}^{\infty} \log\left(1 - \frac{1}{2^{n+1}}\right) \geq -2 \Rightarrow \exp(-2) \leq \prod_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right)$$

Da markov kæden er irreducible har vi i transient $\Rightarrow P$ transient

Derved har vi i transient $\forall i \in S$

- e) *der eksisterer en unik stationærfordeling* $\Leftrightarrow \sum_{i=1}^{\infty} p_0 \cdot \dots \cdot p_i < \infty$
 $\Rightarrow$

Lad π være en stationær fordeling

For en stationær fordeling har vi

$$\pi = \pi P, \quad \sum_{n=0}^{\infty} \pi_n = 1, \quad \pi_i \geq 0$$

Altså for denne markovkæde

$n \geq 1$:

$$\begin{aligned} \pi_0 &= \sum_{i=0}^{\infty} \pi_i (1 - p_i) = 1 - \sum_{i=0}^{\infty} \pi_i p_i = 1 - \sum_{i=0}^{\infty} \pi_{i+1} \\ &= 1 - \sum_{n=1}^{\infty} \pi_n \end{aligned}$$

Rekursivt kan vi skrive

$$1 = \pi_0 + \sum_{n=0}^{\infty} \pi_0 \cdot p_0 \cdot \dots \cdot p_n$$

Per ovenover er rækken monoton og begrænset og derved konvergent

Vi ved også at $\pi_0 \neq 0 \Rightarrow \pi_0 > 0$

$$\Leftrightarrow 1 = \pi_0 \left(1 + \sum_{n=0}^{\infty} p_0 \cdot \dots \cdot p_n\right)$$

Derved

$$\pi_0 = \frac{1}{(1 + \sum_{n=0}^{\infty} p_0 \cdot \dots \cdot p_n)} > 0 \Rightarrow \sum_{n=0}^{\infty} p_0 \cdot \dots \cdot p_n < \infty$$

" $\Leftarrow$ ": vi vil gerne bruge remark 6.24(2) til at sige at det er nok at vise at der findes en stationær fordeling da P er irreducible da den i det tilfælde er unik

Betragt

$$\begin{aligned}\pi_n &= p_{n-1} \cdot \pi_{n-1} \\ \pi_0 &= 1 - \sum_{n=1}^{\infty} \pi_n \\ \pi_0 &= \frac{1}{(1 + \sum_{n=0}^{\infty} p_0 \cdot \dots \cdot p_n)} > 0\end{aligned}$$

Overholder $\pi = \pi P$ per ovenover

Men $p_i > 0$ per antagelse derved

$$\sum_{n=0}^{\infty} \pi_n = \pi_0 + \sum_{n=0}^{\infty} \pi_{n-1} \cdot p_{n-1} > 0 \forall n \in \mathbb{N}$$

$$\sum_{n=0}^{\infty} \pi_n = \pi_0 + \sum_{n=0}^{\infty} \pi_0 \cdot p_0 \cdot \dots \cdot p_n = \pi_0 \left(1 + \sum_{n=0}^{\infty} p_0 \cdot \dots \cdot p_n \right) = 1$$

Per resultatet er vi færdige

Opgave 3

a) Vis $\mathbb{E}_{\mu}[N_i] = \mathbb{P}_{\mu}(T_i < \infty) \cdot (1 + \mathbb{E}_i[N_i])$

Bruger (6.2)

$$\begin{aligned}\mathbb{E}_{\mu}[N_i] &= \mathbb{E}_{\mu}[\mathbb{I}_{\{T_i < \infty\}} \cdot (1 + N_i \circ \theta^{T_i})] \\ &= \mathbb{E}_{\mu}[\mathbb{I}_{\{T_i < \infty\}}] + \mathbb{E}_{\mu}[\mathbb{I}_{\{T_i < \infty\}} \cdot N_i \circ \theta^{T_i}]\end{aligned}$$

1. middelværdi er bare

$$\mathbb{E}_{\mu}[\mathbb{I}_{\{T_i < \infty\}}] = \mathbb{P}_{\mu}(T_i < \infty)$$

Kigger på den 2. middelværdi

$$\begin{aligned}\mathbb{E}_{\mu}[\mathbb{I}_{\{T_i < \infty\}} \cdot N_i \circ \theta^{T_i}] &= \mathbb{E}_{\mu}[E_{\mu}[\mathbb{I}_{\{T_i < \infty\}} \cdot N_i \circ \theta^{T_i} | \mathcal{F}_{T_i}^n]] \\ &= E_{\mu}[\mathbb{I}_{\{T_i < \infty\}} \cdot \mathbb{E}_i[N_i]] = \mathbb{E}_i[N_i] \cdot E_{\mu}[\mathbb{I}_{\{T_i < \infty\}}] = \mathbb{E}_i[N_i] \cdot \mathbb{P}_{\mu}(T_i < \infty)\end{aligned}$$

1. ulighed er torn-egenskaben 2. er den stærke markov egenskab 3. er fordi den indre middelværdi er konstant

Sammen giver

$$= \mathbb{P}_\mu(T_i < \infty) \cdot (1 + \mathbb{E}_i[N_i])$$

b) Vis $\mathbb{P}_\mu(T_i^n < \infty) = 1 \forall n \in \mathbb{N}$

Vi bruger (6.3) fra propositionen

Lad $n \in \mathbb{N}$

$$\mathbb{P}_\mu(T_i^n < \infty) = \mathbb{P}_\mu(T_i < \infty) \cdot [\mathbb{P}_i(T_i < \infty)]^{n-1} = 1 \cdot 1^{n-1} = 1$$

c) Vis at $T_i^1 \perp T_i^2 - T_i^1$ under $\mathbb{P}_\mu$ og at $T_i^2 - T_i^1$ under $\mathbb{P}_\mu$ har samme fordeling som T_i under $\mathbb{P}_i$

$$(6.1) \Leftrightarrow T_i^2 - T_i^1 = T_i \circ \theta^{T_i^1} \text{ under } \{T_i^1 < \infty\}$$

$$\text{Da } \mathbb{P}_\mu(T_i^1 < \infty) = 1$$

Lighed af fordelinger

$$\text{Lad } B \subseteq \mathbb{N} \cup \{\infty\}$$

$$\begin{aligned} \mathbb{P}_\mu(T_i^2 - T_i^1 \in B) &= \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^2 - T_i^1 \in B\}} \right] = \mathbb{E}_\mu \left[\mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^2 - T_i^1 \in B\}} \middle| \mathcal{F}_{T_i^1}^X \right] \right] \\ &= \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^2 - T_i^1 \in B\}} \middle| \mathcal{F}_{T_i^1}^X \right] = \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i \circ \theta^{T_i^1} \in B\}} \middle| \mathcal{F}_{T_i^1}^X \right] \\ &= \mathbb{E}_{X_{T_i^1}} \left[\mathbb{I}_{\{T_i \in B\}} \right] = \mathbb{E}_i \left[\mathbb{I}_{\{T_i \in B\}} \right] = \mathbb{P}_i(T_i \in B) \end{aligned}$$

Derved mangler vi at vise

$$T_i^1 \perp T_i^2 - T_i^1 \text{ under } \mathbb{P}_\mu$$

$$\text{Lad } A, B \subseteq \mathbb{N} \cup \{\infty\}$$

$$\begin{aligned} &\mathbb{P}_\mu(T_i^1 \in A, T_i^2 - T_i^1 \in B) \\ &= \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^1 \in A\}} \cdot \mathbb{I}_{\{T_i^2 - T_i^1 \in B\}} \right] = \mathbb{E}_\mu \left[\mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^1 \in A\}} \cdot \mathbb{I}_{\{T_i^2 - T_i^1 \in B\}} \middle| \mathcal{F}_{T_i^1}^X \right] \right] \\ &= \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^1 \in A\}} \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^2 - T_i^1 \in B\}} \middle| \mathcal{F}_{T_i^1}^X \right] \right] = \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^1 \in A\}} \mathbb{P}_i(T_i \in B) \right] = \mathbb{P}_i(T_i \in B) \cdot \mathbb{E}_\mu \left[\mathbb{I}_{\{T_i^1 \in A\}} \right] \\ &= \mathbb{P}_i(T_i \in B) \cdot \mathbb{P}_\mu(T_i^1 \in A) \\ &= \mathbb{P}_\mu(T_i^2 - T_i^1 \in B) \cdot \mathbb{P}_\mu(T_i^1 \in A) \end{aligned}$$

Derved

$$T_i^1 \perp T_i^2 - T_i^1 \text{ under } \mathbb{P}_\mu$$

Opgave 4

$$P = \begin{pmatrix} 0.1 & 0.4 & 0.1 & 0 & 0 & 0.4 \\ 0 & 0.2 & 0.3 & 0.5 & 0 & 0 \\ 0 & 0 & 0.2 & 0.3 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0.2 & 0.2 & 0.1 \\ 0 & 0 & 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

1, 2 er transiente stadier da

$$\mathbb{P}_1(T_1 < \infty) = 0.1 < 1$$

$$\mathbb{P}_2(T_2 < \infty) = 0.2 < 1$$

1, 2 er transiente, da per t  opgave ved vi at $\pi_1 = \pi_2 = 0$ i en station r fordeling

Hvis vi betragter

$$C = \begin{pmatrix} 0.2 & 0.3 & 0.5 & 0 \\ 0.5 & 0.2 & 0.2 & 0.1 \\ 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Ser vi at den er irreducible da

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 1$$

Og $S \setminus \{1,2\}$ er finite da er C positiv recurrent per 6.23(5)

Derved findes der en unik stationær fordeling π for C

Så lad os finde π

$$\begin{aligned} x &= \frac{4}{5} \cdot z + \frac{1}{2} \cdot y + \frac{1}{5} \cdot x \\ y &= \frac{1}{5} \cdot y + \frac{3}{10} \cdot x \\ z &= k + \frac{1}{5} \cdot z + \frac{1}{5} \cdot y + \frac{1}{2} \cdot x \\ k &= \frac{1}{10} \cdot y \end{aligned}$$

$\Leftrightarrow$

Ligningssystemet løses for x, z, y, k vha. WordMat's 'Løs Ligninger' funktion,

$$k = \frac{12 \cdot z}{245} \quad \wedge \quad x = \frac{64 \cdot z}{49} \quad \wedge \quad y = \frac{24 \cdot z}{49}$$

$$\begin{aligned} x + z + y + k &= 1 \\ \frac{64 \cdot z}{49} + \frac{24 \cdot z}{49} + z + \frac{12 \cdot z}{245} &= 1 \end{aligned}$$

$\Leftrightarrow$

Ligningen løses for z vha. WordMat.

$$\begin{aligned} z &= \frac{245}{697} \\ k &= \frac{12}{245} \cdot \frac{245}{697} = \frac{12}{697} \\ x &= \frac{64}{49} \cdot \frac{245}{697} = \frac{320}{697} \\ y &= \frac{24}{49} \cdot \frac{245}{697} = \frac{120}{697} \end{aligned}$$

Derved er $\begin{pmatrix} 0 & 0 & \frac{320}{697} & \frac{120}{697} & \frac{245}{697} & \frac{12}{697} \end{pmatrix}$ en stationær fordeling for P

Denne er unik da $\pi_1 = \pi_2 = 0$ er tvunget og da kræver vi at $(\pi_3 \quad \pi_4 \quad \pi_5 \quad \pi_6)$ er en stationær fordeling for C og denne er unik må $\begin{pmatrix} 0 & 0 & \frac{320}{697} & \frac{120}{697} & \frac{245}{697} & \frac{12}{697} \end{pmatrix}$ være en unik stationær fordeling for P

$$\pi \text{ stationær fordeling for } P \Leftrightarrow \pi P = \pi, \pi \geq 0, \sum \pi_i = 1$$

Men $\pi_1 = \pi_2 = 0$ og vi ser da at

$$\pi P = \pi, \pi_1 = \pi_2 = 0 \Leftrightarrow (\pi_3 \quad \pi_4 \quad \pi_5 \quad \pi_6) C = (\pi_3 \quad \pi_4 \quad \pi_5 \quad \pi_6), \pi_1 = \pi_2 = 0$$

Hvor $\pi \geq 0, \sum \pi_i = 1$

1,2 transiente

$$\Rightarrow \mathbb{E}_1[T_1] = \mathbb{E}_2[T_2] = \infty$$

Per (6.11) kan vi regne middelværdierne for 3,4,5,6 nemt da de er positivt racerente stadier

$$E_3[T_3] = \frac{1}{\pi_3} = \frac{697}{320}$$

$$E_4[T_4] = \frac{1}{\pi_4} = \frac{697}{120}$$

$$E_5[T_5] = \frac{1}{\pi_5} = \frac{697}{245}$$

$$E_6[T_6] = \frac{1}{\pi_6} = \frac{697}{12}$$

b).

$$\text{se at } \mathbb{P}_1(\tau < \infty) = \mathbb{P}_1(\{T_5 < \infty\} \setminus \{T_5 = 1\}) = \mathbb{P}_1(T_5 < \infty) - p_{15} = \mathbb{P}_1(T_5 < \infty)$$

så vi kan nøjes med at vise

$$\mathbb{P}_1(T_5 < \infty) = 1$$

Definer $\mathcal{T} = \inf\{n \geq 1: X_n \in \{3,4,5,6\}\}$ hvilket er en stoppetid

Da $\{1,2\}$ er transiente hav vi $\mathbb{P}_1(\mathcal{T} < \infty) = 1$ da kan vi skrive

$$\mathbb{P}_1(T_5 < \infty) = \mathbb{E}_1[\mathbb{I}_{\{T_5 < \infty\}}] = \mathbb{E}_1[\mathbb{E}_1[\mathbb{I}_{\{T_5 < \infty\}} | \mathcal{F}_{\mathcal{T}}^X]]$$

$$\mathbb{E}_1[\mathbb{I}_{\{T_5 < \infty\}} | \mathcal{F}_{\mathcal{T}}^X] = E_{X_{\mathcal{T}}}[\mathbb{I}_{\{T_5 < \infty\}}] = \mathbb{P}_{X_{\mathcal{T}}}(T_5 < \infty)$$

Her bruger vi først torn egenskaben og så den stærke markov egenskab

Men $X_{\mathcal{T}} \in \{3,4,5,6\}$ n.s. og da C er positivt recurent er $\mathbb{P}_{X_{\mathcal{T}}}(T_5 < \infty) = 1$

Derved

$$\mathbb{P}_1(T_5 < \infty) = \mathbb{E}_1[1] = 1$$

$$\mathbb{E}_1[X_{\tau+1}] = \mathbb{E}_1[\mathbb{E}_1[X_{\tau+1} | \mathcal{F}_{\tau}^X]] =$$

$$\mathbb{E}_1[X_{\tau+1} | \mathcal{F}_{\tau}^X] = \mathbb{E}_{X_{\tau}}[X_1] = \mathbb{E}_5[X_1] = 0.8 \cdot 3 + 0.2 \cdot 5 = 3.4$$

$$\mathbb{E}_1[X_{\tau+1}] = \mathbb{E}_1[3.4] = 3.4$$